# Badi Ramana

**Mobile Number:** +919347767503
**Email:** ramanabadi83@gmail.com
**LinkedIn:** https://www.linkedin.com/in/ramana-badi-88292a227
**Portfolio:** https://ramana1229.github.io/Ramana/#research
**Location:** Srikakulam, Andhrapradesh, 532407, India

## Professional Summary:

✓ Electrical and Electronics Engineering postgraduate specializing in Advanced Electrical Power Systems, with experience in SAP ABAP development and hands-on exposure to electric-vehicle hardware. Academic project experience includes MATLAB/Simulink modelling of renewable-energy-integrated EV microgrids, power-quality analysis, harmonic mitigation, smart charging, and dual-bus EV power flow. Currently working as a Project Associate-I at NIT Tiruchirappalli, with a research focus on threat modelling and data-driven cybersecurity solutions for networked microgrids.

## Research Experience:

**Project Associate-I, National Institute of Technology Tiruchirappalli (NIT Trichy)** | 2026

- ✓ Research focus: **"Threat Modelling and Data-Driven Cybersecurity Solutions for Networked Microgrids."**
- ✓ Research area connects power-system/microgrid knowledge with cybersecurity and data-driven methods.

## Work Experience & Internship:

- **Capgemini, Kolkata – Software Analyst** (Sept-2022 to 2024)
  - ✓ Worked in corporate environment, gaining teamwork and client-handling experience.
  - ✓ Learned professional communication and project management discipline.
- **ERIDE E-Mobility Company, Hyderabad** (2025)-Hardware Internship
  - ✓ Worked on assembling/disassembling EVs, wiring, and testing BLDC & PMSM motors.
  - ✓ Learned integration of battery, controller, and power distribution systems.
  - ✓ Observed and worked on battery management systems and EV wiring.
  - ✓ Hands-on experience with EV components like motors, controllers, and inverters.

## Academic Projects:

**1. Mitigation of Voltage Drop, Harmonic Distortion and Peak Load Issues in Renewable Energy Integrated EV Microgrid Using MATLAB/Simulink**

- ✓ Developed a renewable-energy-integrated EV microgrid model in MATLAB/Simulink.
- ✓ Implemented V2G/G2V and smart-charging strategies in the model.
- ✓ Designed an LCL filter to reduce harmonics and improve power quality.
- ✓ Performed voltage-profile, THD, and peak-load analysis.
- ✓ Reported THD values of 8.08% without the filter and 4.8% with the filter.

**2. Power Flow Analysis of Dual-Bus Electric Vehicle System with Auxiliary Load Integration Using MATLAB/Simulink**

- ✓ Modeled a dual-bus EV system with battery, DC-DC converter and auxiliary loads.
- ✓ Analyzed power flow, battery SOC, voltage, and current characteristics.
- ✓ Evaluated system performance under different operating conditions.

## Educational Qualification:

- ✓ **M.Tech** – Electrical and Electronics Engineering (Advanced Electrical Power Systems), Jawaharlal Nehru Technological University Kakinada | 2024–2026 | 80%
- ✓ **B.Tech** – Electrical and Electronics Engineering (EEE) Acharya Nagarjuna University, Guntur | 2018–2022 | 78.5%
- ✓ **Intermediate** – MPC
  AP Board of Intermediate | 2016–2018 | 96%
- ✓ **SSC (10th Class)**
  AP Secondary Education | 2016 | GPA: 9.3

## Technical Skills:

- ✓ **Power Systems & Grid:** Power-system analysis fundamentals, renewable-energy integration, microgrids, power quality, voltage-profile analysis, harmonic/THD analysis, smart grids, HVDC, FACTS, protection systems.
- ✓ **EV & Power Electronics:** EV system integration, BLDC/PMSM motors, batteries, motor controllers, inverters, DC–DC converters, auxiliary loads, battery state-of-charge analysis, V2G/G2V concepts.
- ✓ **Simulation & Programming:** MATLAB/Simulink, Powergui, FFT/THD analysis, Python (basics).
- ✓ **Software Development:** SAP ABAP, classical and interactive reports, ALV reports, Data Dictionary (DDIC), SmartForms, Adobe Forms, function modules, enhancements (User Exits/BAdIs), ABAP Objects (OOP).
- ✓ **Professional:** Problem solving, communication, teamwork, analytical thinking, adaptability, documentation and presentation.

## Research Interests:

- ✓ Power System Analysis, Renewable Energy Systems, Electric Vehicle Technology, Smart Grid, Cybersecurity
- ✓ Power Electronics, Battery Energy Storage Systems, Grid Integration of Renewable Energy

## Declaration:

I hereby declare that the above information is true and correct to the best of my knowledge and belief.

**Place:** _____

**Date:** _____

**Ramana Badi**